

A Bio-sensing System for Visualizing Material Perceptions through Physiological Computing and Geometric Encoding

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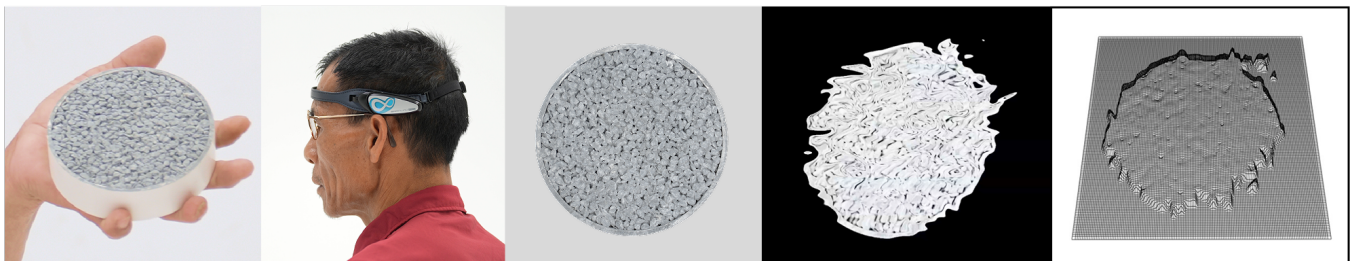


Figure 1: System flow overview. From tangible material perception to cognitive mapping through physiological computing and geometric encoding (from left to right).

Abstract

We propose a bio-sensing system that translates material perceptions into geometric encoding. Brain activity frequencies and post-filtered real-time Electroencephalography (EEG) data are mapped to turbulence, brightness, and particle dispersion, generating parameterized forms. This approach explores how physiological computing can inform scientific visualization through geometric encoding, opening new directions for linking EEG biosensing, computer graphics, and material perceptions.

CCS Concepts

• **Human-centered computing** → **Visualization design and evaluation methods**; *Information visualization*; *Visualization techniques*.

Keywords

Geometric Encoding, EEG Biosensing, Physiological Computing, Cognitive Mapping, Material Perception

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SA Posters '25, Hong Kong, Hong Kong
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ACM ISBN 979-8-4007-2134-2/25/12
<https://doi.org/10.1145/3757374.3771478>

ACM Reference Format:

Mirna Zordan, Enza Migliore, Minghong Zhang, and Peiyue Lin. 2025. A Bio-sensing System for Visualizing Material Perceptions through Physiological Computing and Geometric Encoding. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757374.3771478>

1 Introduction

Quantifying human responses to construction materials requires instrumentation that links physiological signals to interpretable visual and geometric representations. Prior work in physiological computing shows that EEG band-power features (α , β , θ , γ) and derived indices can track task-related cognitive states [Fairclough 2009; Pope et al. 1995], yet tools that couple these features with real-time visualization and geometric encoding for material studies remain limited. We present a compact, reproducible system that acquires EEG data via a wearable headset (BrainLink), performs real-time preprocessing, extracts band-power and composite features, and streams them to a real-time renderer in TouchDesigner. Features are mapped to parametric behaviors, and processed data are then encoded to explore future applications in digital fabrication processes.

2 Related Work

Physiological computing has established EEG band-power and composite indices as proxies for workload and engagement, enabling signal-driven interfaces [Fairclough 2009; Pope et al. 1995]. In design and HCI, materials are recognized as shaping perception and

interaction; frameworks of material experience argue that perceptual attributes (e.g., roughness, reflectiveness) inform user experience and design decisions [Giaccardi and Karana 2015]. Scholars in the field of the built environment further situate concrete as a consequential medium in urban life, extending beyond engineering performance to broader implications for practice. Our work connects these strands by translating EEG biosensing into real-time visual and geometric encoding for material studies; we start from concrete as a testing material, providing a reproducible system tailored to future potentials in digital fabrication, accounting physiological computing.

3 System Design

We developed an end-to-end workflow that acquires EEG signals in real time, performs online preprocessing, and transforms the extracted EEG features into parameterized visual and geometric encodings through cognitive mapping. We use Bluetooth (serial COM port) to connect to an EEG sensor (BrainLink), and we apply a Kalman Filter to smooth the data and remove jitter. Processed results are sent via UDP to TouchDesigner, with different features assigned to specific ports: stress (5000), engagement (5003), and relaxation through α band power (5002). In cases of signal loss or poor quality, the system used the last received value or a default fallback (Signal = 1). For further computing of physiological data, we mapped EEG frequencies to represent three cognitive states for initial exploration and testing (Figure 2), allows for real-time interaction and digital manipulation of the material:

- (1) Stress [Jebelli et al. 2018]: we extracted the stress index and mapped data to the *Noise* operator, where the *Exponent* parameter controls the surface perturbation intensity. Higher stress values produced increasingly turbulent textures.
- (2) Engagement [Pope et al. 1995]: we extracted the engagement index and mapped data to a GPU-based particle system (*Particle LesGPU*), where the engagement index modulates particle dispersion and clustering. Higher engagement led to more dispersed spatial distributions.
- (3) Relaxation [Risqiawati et al. 2024]: we used the power of high alpha waves directly to represent the relaxation state of the brain; here, α band power is mapped to the *Level* operator, which controls the degree of distortion. Higher α values resulted in more luminous and distorted geometries.



Figure 2: Physiological computing of three cognitive states: stress, engagement, relaxation (from left to right).

Subsequently, we imported selected images of real-time visual effects into Rhinoceros and transformed them into geometric encodings through Grasshopper. The images were sampled over a

rectangular point grid, where their grayscale values were mapped into numerical data and translated into displacement factors along the Z axis. Driven by these displacements, the point grid formed a spatial point set with varying elevations, which was then built into a continuous editable surface (Figure 3).

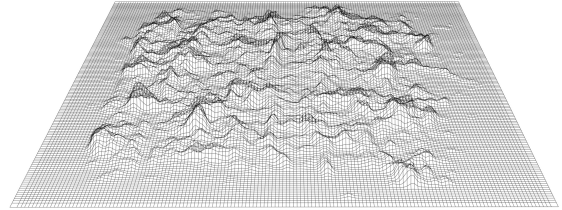


Figure 3: Relaxation. Geometric encoding from physiological computing.

Parameter traces from the renderer are logged and can be reused to regenerate both visualizations and digital fabrication meshes, ensuring consistency across modalities.

4 Conclusion and Future Work

We present a reproducible biosensing system that merges physiological computing and geometric encoding for material perception. The system translates real-time EEG data into dynamic geometric encoding that reveals fluctuations in surface turbulence, making otherwise abstract physiological states perceptible as evolving textures. Future work will incorporate more metrics to evaluate stability and extensibility, and will validate its effectiveness through experiments and user studies. Additionally, we will introduce a calibration stage to enhance the system's adaptability for diverse users, and we will deepen visualization criteria. This work sets grounds for future exploration in digital fabrication based on physiological computing. It contributes to the continuity between ephemeral signal visualization and tangible material expression.

Acknowledgments

The authors thank Design Trust Seed Grant for supporting this work, and Wang Yao for her assistance in production.

References

- Stephen H. Fairclough. 2009. Fundamentals of physiological computing. *Interacting with Computers* 21, 1-2 (2009), 133–145. doi:10.1016/j.intcom.2008.10.011
- E. Giaccardi and E. Karana. 2015. Foundations of Materials Experience: An Approach for HCI. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 2447–2456. doi:10.1145/2702123.2702337
- Houtan Jebelli, Sungjoo Hwang, and SangHyun Lee. 2018. EEG-based workers' stress recognition at construction sites. *Automation in Construction* 93 (2018), 315–324. doi:10.1016/j.autcon.2018.05.027
- Alan T. Pope, Edward H. Bogart, and Debra S. Bartolome. 1995. Biocybernetic system evaluates indices of operator engagement in automated task. *Biological Psychology* 40, 1-2 (1995), 187–195. doi:10.1016/0301-0511(95)05116-3
- Diah Risqiawati, Adhi Dharma Wibawa, Evi Septiana Pane, Eko Mulyanto Yuniarno, Wardah Rahmatul Islamiyah, and Mauridhi Hery Purnomo. 2024. Effective relax acquisition: a novel approach to classify relaxed state in alpha band EEG-based transformation. *Brain Informatics* 11, 1 (May 2024). doi:10.1186/s40708-024-00225-y